

Threat Score & Risk Ranges

How Catchers AI turns ML predictions and other checks into the **threat score** (0–100) and risk labels (**LOW**, **MEDIUM**, **HIGH**, **CRITICAL**).

What you see in the UI (main threat score)

The gauge and badges use a **combined threatScore from 0–100**, not the raw ML percentage alone. That score is built from VirusTotal, Google Safe Browsing, PhishTank, ML, HTTPS checks, heuristics, and more, then capped at 100.

Risk bands (final threat score)

Risk label	Threat score range	Meaning
LOW	0 – 24	Less harmful — generally safer with normal precautions
MEDIUM	25 – 49	Some risk — review before proceeding
HIGH	50 – 79	Significant risk — avoid if possible
CRITICAL	80 – 100	Highest risk — do not visit / quarantine

Boundaries are **inclusive** on the upper band (e.g. exactly **50** is **HIGH**, exactly **25** is **MEDIUM**).

Backend (`backend/src/services/threatAnalysis.ts`):

```
if (threatScore >= 80) {  
  riskCategory = 'CRITICAL';  
} else if (threatScore >= 50) {  
  riskCategory = 'HIGH';  
} else if (threatScore >= 25) {  
  riskCategory = 'MEDIUM';  
} else {  
  riskCategory = 'LOW';  
}
```

Frontend (`frontend/src/lib/risk.ts` — `categoryFromScore`):

```
if (score >= 80) return "CRITICAL";  
if (score >= 50) return "HIGH";  
if (score >= 25) return "MEDIUM";  
return "LOW";
```

File scans use the same thresholds in `threatAnalysis.ts`.

What the ML service outputs

The Python ML service does **not** assign LOW / MEDIUM / HIGH / CRITICAL. It returns probabilities and a 0–100 ML score.

Trained model path (Random Forest)

From `predict_proba` in `ml-service/app/ml_engine.py`:

Field	Description
<code>threat_probability</code>	Chance the URL/content is a threat (0.0–1.0, often shown as 0–100%)
<code>safe_probability</code>	Chance it is safe
<code>ml_score</code>	<code>int(threat_probability × 100)</code> → 0–100
<code>confidence</code>	<code>max(safe_probability, threat_probability)</code> — how sure the model is either way , not “how malicious”
<code>is_threat</code>	<code>predict() == 1</code> — sklearn’s default ~50% decision boundary

```
# probabilities[0] = safe, probabilities[1] = threat
safe_prob = float(probabilities[0])
threat_prob = float(probabilities[1])

is_threat = bool(prediction == 1)
confidence = max(safe_prob, threat_prob)
ml_score = int(threat_prob * 100)
```

ML-only binary split (not the same as UI bands)

ML <code>threat_probability</code> / <code>ml_score</code>	Model label
0% – ~49%	Usually <code>is_threat: false</code> (safe)
~50% – 100%	Usually <code>is_threat: true</code> (threat)

There is **no** separate ML mapping for medium vs high vs critical — only safe vs threat at ~50%.

Rule-based fallback (if the `.pk1` model is missing)

A weighted rule score 0–100 is used; `is_threat` if **score ≥ 50**:

```
threat_prob = score / max_score
is_threat = score >= 50
```

How ML feeds the final threat score

ML only **adds** to `threatScore` when the model says `is_threat: true` (`backend/src/services/threatAnalysis.ts`):

```
if (mlResult.prediction.is_threat) {  
  threatScore += Math.round(mlScore * mlConfidence);  
}
```

- **mlScore** — 0–100 from the ML service
- **mlConfidence** — 0–1 (**confidence** from the model)

Example: `ml_score = 80`, `confidence = 0.9` → adds **72** points to the combined score (before cap at 100).

If `is_threat` is false, ML does **not** increase the score; it may record a “classified as safe” security note instead.

Other engines can add large amounts independently, for example:

- VirusTotal — up to +70
- Google Safe Browsing — up to +60
- PhishTank — up to +50
- HTTP without TLS — +25
- Heuristics, shorteners, suspicious TLDs — additional points

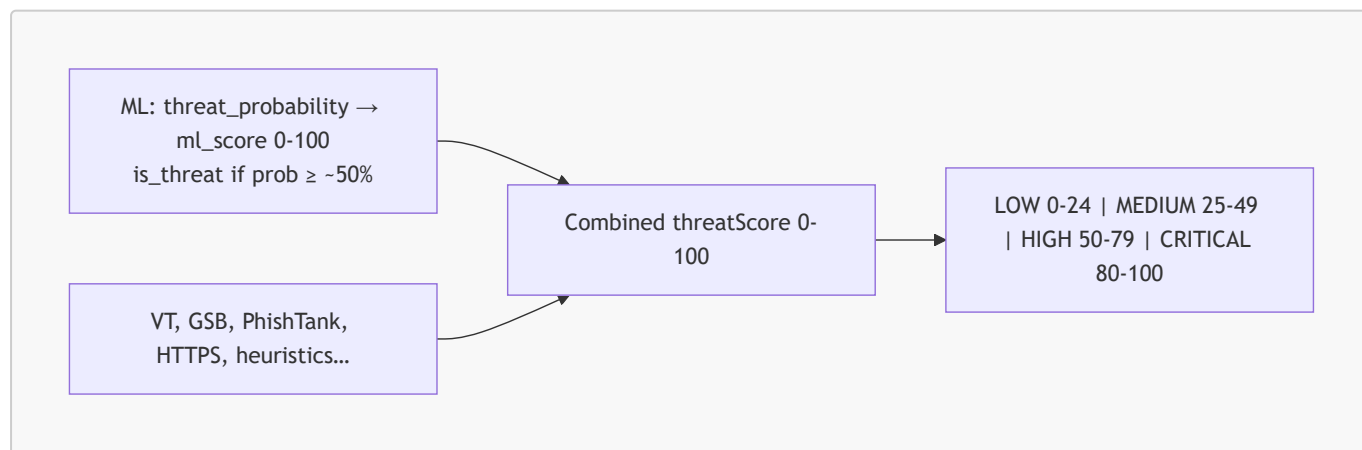
So a **low ML %** can still yield **HIGH** or **CRITICAL** overall, and a **high ML %** might stay **MEDIUM** if little else triggers.

ML confidence labels (informational only)

These appear in scan details from `ml-service/app/feature_extractor.py`. They are **not** risk tiers.

Model confidence	Message
> 90%	“Very high model confidence (>90%)”
> 75%	“High model confidence (>75%)”

Flow overview



Quick reference

Layer	Ranges
UI risk labels	LOW 0–24 , MEDIUM 25–49 , HIGH 50–79 , CRITICAL 80–100 (combined <code>threatScore</code>)
ML threat probability	~ 0–49% → usually safe; ~ 50–100% → usually threat (<code>is_threat</code>)
ML → score contribution	Only when <code>is_threat</code> : adds <code>round(ml_score × confidence)</code>

Source files:

- `backend/src/services/threatAnalysis.ts` — scoring and category assignment
- `frontend/src/lib/risk.ts` — `categoryFromScore`
- `ml-service/app/ml_engine.py` — ML probabilities and `ml_score`
- `ml-service/app/feature_extractor.py` — confidence factor messages